

Licensing Barriers to Interstate Mobility in the US: Evidence from Travel Nurses

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October 5, 2026

Abstract

This paper examines the impact of licensing barriers to interstate mobility on nursing labor markets and health outcomes during the COVID-19 pandemic. Using county-level CDC data, I exploit spatial variation from the Nurse Licensure Compact (NLC) and temporal variation from federal emergency waivers to identify the causal effects of licensing barriers on all-cause mortality. Estimates from a boundary discontinuity design and difference in difference approach show that non-NLC states experienced 6% higher deaths per capita during the first half of 2020, despite having adopted emergency measures intended to ease licensing barriers. To establish the mechanism underlying this effect, I estimate the effect of licensing barriers on nurse decisions using granular data on US travel nurses and their individual licenses. The results reveal that a lack of licensure required a wage premium of 97% for a nurse to be willing to apply to a county. The required wage premium fell to 38% while the emergency measures were in effect but rose to 53% after they expired. Counterfactual estimates reveal that non-NLC states suffered the highest loss in applications due to licensing barriers. Together, the findings highlight that interstate licensing barriers constrained the rapid reallocation of travel nurses during the pandemic and that uniform licensing policies like the NLC can enhance labor market responsiveness and improve public health resilience in times of crisis.

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1 Introduction

Occupational licensing requires workers to obtain state authorization to practice. A key feature of licensing policies in the US is that they vary across states, creating barriers to interstate migration ([Johnson and Kleiner, 2020](#)). In healthcare, the largest licensed sector of the US workforce and one facing persistent staffing shortages, such barriers can affect patient outcomes, especially during emergencies.¹ Yet little is known about how interstate licensing barriers affect the mobility of healthcare workers and patient outcomes. This paper bridges this gap by examining how interstate licensing variations shaped the labor market for Registered Nurses (RNs) during COVID-19 and affected mortality across states.

A growing body of evidence links nurse staffing to patient mortality across settings ([Propper and Van Reenen, 2010](#); [Gruber and Kleiner, 2012](#); [Cook et al., 2012](#); [Friedrich and Hackmann, 2021](#)). Notably, ([Gruber and Kleiner, 2012](#)) find that nurse strikes in New York State raised in-hospital mortality by 18%. Public health emergencies can intensify this effect by sharply increasing demand in a hospital system that already reports an average RN vacancy rate of 8%. During COVID-19, hospitals responded to this gap by increasingly relying on travel nurses - RNs hired for temporary assignments - to meet the sudden surge in demand.² ³ Because these assignments often require nurses to cross state lines, state-specific licensing requirements can impede their movement to areas of greatest need. This paper focuses on travel nurses, and their licensing process is the same as that of a regular nurse.

All Registered Nurses (RNs) in the US must hold a license to practice in any state, with licenses not necessarily portable across states. However, for RNs, there exists an agreement - Nurse Licensure Compact (NLC) - that allows them to practice in any member state under a single “multi-state” license. This eliminates the need for relicensure when moving between member states. By 2020, 30 states were members (NLC states), while 20 were not (non-NLC states). This could have placed the non-NLC states at a disadvantage when responding to the sudden increase in demand for nurses. However, one other feature of the licensing policies during COVID-19 was the adoption of federal emergency waivers.

During the federal public health emergency, from March 2020 to May 2023, states temporarily eased mobility barriers by suspending interstate licensing requirements. Although

¹“The U.S. health workforce continues to struggle with staffing shortages and maldistribution of health professionals” ([American Hospital Association, 2023](#)).

²[U.S. Government Accountability Office \(2024\)](#).

³As per data from Aya Healthcare, one of the leading healthcare providers in the US, demand for travel nurses jumped by 217% in November 2020 from a year before.

all states adopted waivers in March 2020, implementation details emerged only gradually through state-specific licensing memoranda. This uncertainty likely delayed nurses' response to the emergency measures, leaving non-NLC states still reliant on nurses already licensed within the state during the initial COVID-19 outbreak.⁴ Beyond quantifying the effects of licensing barriers on interstate mobility, this setting also allows to evaluate how effectively the emergency measures eased these barriers.

This paper combines novel data on travel nurse applications from 2019-2023 with the mortality data from CDC Wonder from 2018-2024 to study the effects of interstate licensing barriers on travel nurse labor market and mortality. In particular, this paper addresses three research questions: First, did mortality differ between NLC and NonNLC states in ways that can be attributed to differences in licensing barriers for nurses? Second, to what extent do licensing barriers affect nurses' application decisions, beyond any effect operating through wages? Third, to what extent these differences in applications due to licensing barrier can explain the mortality differences obtained in the first part?

The empirical analysis proceeds in three steps. First, to estimate the effect of state NLC participation on all-cause mortality, I compare county-level deaths per capita from 2018 to 2024. Because NLC and non-NLC states differed along several dimensions, I use a boundary discontinuity design (BDD) with difference in difference approach. I compare neighboring counties on opposite sides of state borders, where one county is in an NLC state and the other is in a NonNLC state. This reduces observable differences between the two groups, including differences in COVID-19 caseload. Further, I also include half-yearly interactions to examine how the difference changed over time. The results show that, despite emergency measures already being in place, the gap between non-NLC and NLC deaths per capita increased by 6% in 2020H1, compared to the 2019H2 baseline. It moved back to the 2019H2 levels in 2020H2, and continued to remain so.

These results provide evidence that emergency measures eventually narrowed the gap between NLC and non-NLC states, but only with a lag. During the initial COVID-19 outbreak, non-NLC states experienced 6% higher deaths per capita, attributable to their higher interstate licensing barriers. Because the BDD estimates may still capture other state-level policy differences that persist at state borders, I complement the reduced-form analysis with a model of travel nurse applications and hospitals' wage-posting decisions.

In the next part of the analysis, the model uses rich travel nurse applications data

⁴Figure A1 in the Appendix provides anecdotal evidence of the delay in response to emergency waivers.

obtained from Wanderly between 2019 and 2023.⁵ I observe every application submitted through the platform, the characteristics of each job posting, and each nurse’s home state and license portfolio. Crucially, nurses can apply to positions in states where they are not yet licensed and complete the licensing process after receiving an offer, typically with assistance from the staffing agency. The data therefore contain both licensed and unlicensed applications. Variation in whether a nurse is authorized to practice in each destination state provides the identifying variation for estimating the effect of licensing barriers on application decisions. For unlicensed applications, the primary remaining cost is the delay associated with obtaining a new license. Using information on the NLC status of the home state, I also account for selection into licensing.

Following [Nevo \(2001\)](#), I first estimate a supply-side model of travel-nurse applications and then use the estimated parameters to recover county-level wage markdowns. The model identifies how interstate licensing barriers affect application decisions, conditional on distance and county-level mean utility. I decompose mean utility into posted wages and COVID-19 cases, instrumenting for wages in a 2SLS specification to estimate the wage elasticity of applications. The results show that licensing barriers deter cross-state applications. In 2019, nurses required a wage premium as high as 97% to apply to a destination where they were not licensed. This willingness to accept fell to 38% under the emergency measures but rose to 53% as the measures began to expire. Consistent with the fact that emergency measures were temporary and unclear, we do not see a full closing of the licensing penalty.

In the final step, I use the model to simulate the removal of licensing barriers by allowing every nurse to practice in every state. The largest application gains accrue to non-NLC states.⁶ NLC states also experience counterfactual gains through 2020H1, indicating that licensing frictions persisted until 2020H2, consistent with the lag observed in the reduced-form results. I then use the counterfactual increase in applications as a county-level measure of exposure to licensing barriers. Counties with greater exposure experienced 8% higher deaths per capita in 2020H1, but this gap subsequently closed. Together, these results show that licensing barriers constrained travel nurse applications and increased mortality until nurses adjusted to the emergency measures.

⁵Wanderly is a digital staffing marketplace for travel nurses where they can compare across job postings from multiple agencies and choose where to apply. Once the application is made, the final hiring process is handled by the agency and not Wanderly. Because of this, I only see applications in my data and not the final match.

⁶The model uses nurses’ license portfolios and does not directly incorporate the NLC status of destination states.

To the best of my knowledge, this is the first paper to provide evidence on the mortality effects of interstate licensing barriers in the United States. I show that although emergency measures helped mitigate these barriers during COVID-19, they did so only with a lag; during this adjustment period, constraints on nursing supply resulted in higher mortality. Temporarily suspending state-specific licensing requirements is a common response to emergencies such as wildfires and hurricanes. Yet little is known about the consequences of relying on temporary emergency waivers rather than removing interstate licensing barriers in advance. I provide evidence of the lag in adjustment to these waivers.

This paper contributes to the growing literature on occupational licensing. Most existing work studies licensing as a barrier to entry into an occupation (Kleiner and Soltas, 2023; Kleiner and Krueger, 2013; Blair and Chung, 2019). Fewer papers examine licensing as a barrier to mobility across states (Johnson and Kleiner, 2020; Bae and Timmons, 2023), and studies of the NLC find limited effects on nurse supply or migration (DePasquale and Stange, 2016; Shakya et al., 2022). By linking individual licensing records to job applications and posted wages, this paper directly identifies how interstate licensing barriers affect nurses' application decisions and the wages associated with those barriers. Moreover, it focuses on travel nurses, for whom interstate licensing requirements are especially likely to bind.

This paper also contributes to the literature examining how licensing-induced changes in labor supply affect consumer outcomes. Anderson et al. (2020) show that early midwifery licensing laws reduced maternal mortality in the United States. Atal et al. (2026) study physician licensing in Chile and disentangle its effects on provider quality and quantity, as well as the resulting consequences for mortality. I show that interstate licensing barriers constrained the reallocation of nurses during COVID-19, thereby increasing mortality in non-NLC states.

Finally, this paper contributes to the study of the role of nurses in COVID-19. In particular, Gottlieb and Zenilman (2026) study of the market for travel nurses during COVID-19 closely parallels the focus of this paper. They find that a unified national market for travel nurses facilitated their redistribution to high-demand areas during the pandemic. By establishing the relevance of NLC in explaining differences in COVID-19 outcomes of states, I provide evidence that variation in nurse licensure across some states inhibited the integration of this national market for nurses at least in the initial months of the pandemic outbreak.

The rest of the paper is structured as follows. Section 2 provides the institutional background. Section 3 describes the data used for the analysis. Empirical strategy and reduced-

form evidence are discussed in section 4. Section 5 discusses in detail the structural equilibrium model. Parameter results and counterfactuals are discussed in section 6 I conclude in section 7.

2 Institutional Details

2.1 Registered Nurse Licensure in the US

The US nursing labor market is broadly divided across four professions: Certified Nursing Assistant (CNA), Licensed Practical Nurse (LPN), Registered Nurse (RN), and Advanced Practice Registered Nurse (APRN). This paper focuses on RNs, whose primary responsibilities include assessing patients, making care plans, giving treatments, and coordinating care. All RNs in the US must hold a state-issued license to practice, and they make up about 77% of the licensed nurses in the US. To obtain an initial license, an RN must graduate from an accredited nursing program, and pass the national licensing exam (NCLEX-RN), after which they can apply through a state’s board of nursing.

To practice in another state, an RN could be required to obtain the state’s license through “licensure by endorsement”. Under this process, the nurse does not have to give the licensing exam again but still must complete background checks and other administrative requirements. Although these steps are largely bureaucratic, the time needed to obtain a new license ranges from 10 days to 8-12 weeks across states. Renewal requirements also differ considerably across states, including continuing education requirements and how often a license must be renewed. Together, these differences can make it costly for nurses to move or work across state lines. With an ageing population and growing RN shortages, these barriers can limit how quickly the RN workforce adjusts to shifts in demand across states. To address this issue, the National Council of State Board of Nursing (NCSBN) introduced the Nurse Licensure Compact (NLC) in 2000. The NLC allows a nurse to practice in any member state with a single multistate license, removing the licensing barrier to interstate mobility of RNs. Texas, Wisconsin, and Utah were the first three states to pass the NLC into law in 2000, and by the onset of COVID-19 in 2020, 30 states were a part of the compact (*NLC States*) and 20 were not (*Non-NLC States*). Figure 1 maps these states and Figure 2 shows the distribution of the time and fee cost of obtaining a new RN license by the NLC status of the state.

NLC Status

- Non-NLC
- NLC

Figure 2: Time and Cost of Obtaining a New RN License by NLC Status

A stacked bar chart showing the number of states for Non-NLC (blue) and NLC (green) across four time intervals: 2-4 weeks, 4-6 weeks, 6-8 weeks, and 8-12 weeks. The y-axis represents the 'Number of states' from 0 to 25. The legend indicates that blue represents Non-NLC and green represents NLC.

Time Interval	Non-NLC	NLC
2-4 weeks	0	14
4-6 weeks	10	17
6-8 weeks	7	0
8-12 weeks	1	0

State	Non-NLC Fee (\$)	NLC Fee (\$)
CA	300	
MA	225	
MI	205	
NJ	200	
VA	190	185
DC	185	
DE	180	170
AZ	170	160
OK	160	155
NM	155	150
VT	150	
ND	145	140
NY	140	135
WY	135	130
RI	130	
FL	125	120
IL	120	115
KY	115	110
NE	110	105
MT	105	100
WA	100	100
MN	100	
AL	100	100
AR	100	100
KS	100	100
LA	100	100
MS	100	100
MO	100	100
MT	100	100
NV	100	100
SD	100	100
TN	100	100
TX	100	100
WA	100	100
WV	100	100
IA	100	100
SC	100	100
CO	100	100
OK	100	100
GA	100	100
ME	100	100
NH	100	100
OH	100	100
WI	100	100
UT	100	100
MO	100	100

6

Travel Nurses: Another feature of the RN labor market that has grown with rising shortages is the market for travel nurses, which is also the focus of this paper. Travel nurses are hired by hospitals on short-term (typically 3 month) contracts, and the hiring is mediated through a travel nurse agency. The data for this paper is sourced from Wanderly which is a travel nurse job postings aggregator that lists postings from multiple travel nurse agencies in one place for a nurse to compare and apply to. Aside from contract structure, their licensing process and job responsibilities are the same as those of regular nurses. Because they often work across multiple states, licensing barriers can pose greater mobility constraints for them. Hospitals relied heavily on travel nurses during COVID-19 to meet the sudden surge in demand. Figure A2 plots the yearly unfilled RN vacancies from 2018-2023 against the travel nurse assignments. As the vacancy rate went up during COVID-19, so did the travel nurse assignments.

2.2 Emergency Measures and COVID-19

The US has long used emergency declarations to coordinate disaster responses through the Emergency Management Assistance Compact (EMAC), established in 1996 and adopted by all states by 2020. When a governor declares an emergency, EMAC enables states to share personnel and resources, temporarily waiving licensing restrictions for out-of-state workers.

In response to COVID-19, multiple federal and state emergencies were declared. The first, on January 31, 2020, designated COVID-19 as a Public Health Emergency (PHE), granting governors flexibility to request federal assistance but not removing interstate licensing barriers. By March 13, 2020, 13 NLC and 8 NonNLC states had issued their own licensing waivers, though the measures varied widely. That day, President Trump declared a national emergency under the National Emergencies Act, temporarily suspending interstate licensing variation and allowing healthcare workers with valid state licenses to practice nationwide.

The first confirmed US case of COVID-19 was reported on January 20, 2020, and by March 13, over 30,000 cases had been recorded nationwide. The rapid escalation in infections created an urgent demand for healthcare personnel. NLC states were comparatively better positioned to meet this surge, as they could redeploy nurses across member states under the multistate license. In contrast, NonNLC states were constrained by in-state nurse supply until emergency measures were enacted.

3 Data

This paper uses data from three sources: county-level mortality and COVID-19 data from the Centers for Disease Control and Prevention (CDC), and county-level demographic data from the Census Bureau. The data on travel nurse applications and job postings are provided by Wanderly,⁷ an online staffing marketplace that connects travel nurses to short-term hospital contracts through staffing agencies.

The analysis uses monthly county-level all-cause deaths from 2018 to 2024, obtained from the Multiple Cause of Death files in CDC WONDER. These data include mortality by age, race, education, place of death, and cause of death, which I use for robustness checks. The main specification uses all-cause mortality as the outcome. Data on COVID-19 cases and vaccination rates from 2020 to 2023 are obtained from the CDC COVID-19 data supplement. To isolate the effect of nurse licensing barriers on mortality, I control for demographic differences across counties using Census Bureau data on annual county-level distributions by education, age, median household income, urban population share, race, and ethnicity.

Table 1 summarizes demographic characteristics, mortality rates, and COVID-19 measures for counties in NLC and non-NLC states in the full sample. Panel B shows that non-NLC states reported more deaths on average but fewer deaths per capita. They also had fewer cases per capita and higher vaccination rates. Moreover, as Panel A shows, counties in NLC and non-NLC states differed considerably across several demographic characteristics. Thus, comparing mortality across the two groups - even while exploiting temporal variation in emergency licensing waivers - is insufficient to isolate the effect of interstate licensing barriers on mortality.

To account for these differences, the first part of the empirical analysis (Section 4) adopts a boundary discontinuity design (BDD). The design compares counties near opposite sides of state borders separating NLC and non-NLC states, thereby substantially narrowing the observable differences between the two groups. Section 4.1 examines the resulting covariate balance in detail. Although the BDD accounts for many observable differences, unobserved differences and state policies correlated with NLC status may still confound the estimates. Therefore, to provide more direct evidence on the mechanism through which licensing barriers affect the allocation of nurses, I estimate an equilibrium model of travel nurse applications and wages using data from Wanderly.

⁷Wanderly website. Please refer to Figure A4 for a snapshot of the Wanderly website.

The Wanderly data provides information on all the travel nurse applications made through this portal from 2019 to 2023. After applications are submitted through Wanderly, the final recruitment process is handled by the staffing agency through which the nurse applied. As a result, I observe applications but not final matches or other outcomes of those applications. Nevertheless, observing applications allows me to capture a key margin of nurse decision-making: whether licensing barriers deter nurses from applying to jobs in states where they are not already licensed.

Importantly, the Wanderly data include the set of state licenses held by a travel nurse who submitted an application through the platform. This information allows me to classify the licensing barriers faced by each nurse with greater precision. In addition to license holdings, I observe each nurse’s home state and the job posting to which an application was made. For each job posting, I observe the wages, hospital zipcode, specialty, agency and profession. Wanderly data has information on two professions: Registered Nurses (RNs) and Allied Nurses.

Table 2 presents summary statistics for the Wanderly data, which include 59,535 applications submitted by 9,966 travel nurses between 2019H2 and 2023H2. The average travel nurse wage during the sample period was \$94, although wages varied considerably across the 1,427 counties and five years covered by the data, during which the travel nurse market changed substantially. Importantly, the data report all state licenses held by each nurse, allowing for a precise measure of the licensing barriers each nurse faces. This information is summarized in Panel C of Table 2.

The first section of Panel C shows the number of licenses held by nurses based on their home state’s NLC status. Nurses from NLC states hold an average of 19 licenses, compared with 4 among nurses from NonNLC states. The next section of Panel C show that 39 percent of applications are within the nurse’s home state, 32 percent are to out-of-state jobs where the nurse is licensed, and 29 percent are to out-of-state jobs where the nurse is not licensed. The final section reports the same distribution by the NLC status of the nurse’s home state and the state of the job application, rather than by the nurse’s actual licenses. The similar pattern suggests meaningful differences in licensing barriers between NLC and NonNLC states.

Table 1: Mortality Data Summary Statistics by NLC Status

Panel A: Demographic Distribution			
	Non-NLC	NLC	Difference (Non-NLC – NLC)
Population	226,026	100,397	125,629***
Urban Population Percentage (%)	51.1	40.9	10.2***
Age Share: 0 to 44 (%)	53.8	55.0	-1.2***
Age Share: 45 to 64 (%)	26.1	25.6	0.5***
Age Share: 65 to 74 (%)	11.8	11.4	0.3***
Age Share: 75+ (%)	8.3	8.0	0.3***
Share: Non-Hispanic White (%)	80.4	73.1	7.3***
Share: Non-Hispanic Black (%)	4.5	12.1	-7.6***
Share: Non-Hispanic Asian (%)	2.6	1.4	1.2***
Share: Hispanic (%)	9.4	9.9	-0.5***
Share: Bachelor’s Degree or Higher (%)	27.1	23.7	3.4***
Unemployment Rate (%)	5.1	4.2	0.8***
Median Household Income (\$)	70,264	61,750	8,513***
Panel B: Mortality and Health Indicators			
	Non-NLC	NLC	Difference (Non-NLC – NLC)
Total Deaths	170.6	81.7	88.9***
Deaths Per 100k Population	96.6	107.4	-10.8***
COVID-19 Deaths	15.7	7.0	8.7***
COVID-19 Deaths Per 100k	7.9	9.8	-2.0***
COVID-19 Confirmed Cases	1,505.4	640.3	865.1***
COVID-19 Cases Per 100k	647.9	683.3	-35.4***
Vaccine: Received At Least 1 Dose (%)	32.6	28.3	4.3***
Vaccine: Completed Primary Series (%)	57.0	52.7	4.3***
Vaccine: Received Up-to-Date Series (%)	29.2	24.6	4.6***
County-months, 2018–2024	52,538	154,610	
County-months, 2020–2023 (COVID-19 rows)	28,476	91,284	

Notes: Means over county-months, 2018–2024; COVID-19 and vaccination variables cover 2020–2023. NLC status is the state’s status in the year of the observation. Difference is the Non-NLC mean minus the NLC mean; stars from a two-sample Welch t -test: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Wanderly Data Summary Statistics

Panel A: Application Distributions						
	Count	Mean	Min	Max	Std Dev	N
Year	5	11,907	1,422	23,241	8,193	59,535
Nurses	9,966	6	1	154	9	59,535
Specialty	20	2,977	1	25,906	5,985	59,535
Home State	49	1,215	53	9,868	1,845	59,535
Applied State	49	1,215	44	12,132	1,933	59,535
Applied County	1,427	42	1	2,961	127	59,535
Hospital Zipcode Applied	3,458	17	1	1,136	36	59,535
Panel B: Compensation and Geographic Proximity						
	Count	Mean	Min	Max	Std Dev	N
Loaded Pay (\$)		95	18	193	26	59,535
Base Pay (\$)		58	11	153	26	59,535
Real Loaded Pay (RPP Adjusted \$)		94	18	204	26	59,535
Real Base Pay (RPP Adjusted \$)		57	11	173	27	59,535
Distance (km)		1,193	0	4,302	1,212	59,535
Panel C: Licensure and NLC Status						
	Count	Mean	Min	Max	Std Dev	N
Number of Licenses Held Per Nurse						
Whole Sample		15	1	38	14	9,966
Home State - NLC		19	1	38	14	7,214
Home State - NonNLC		4	1	37	7	2,808
Percent applications by a nurse - by Licensed Status						
Within Home State (%)		39	0	100	43	9,966
Outside Home State - Licensed (%)		32	0	100	40	9,966
Outside Home State - Non-Licensed (%)		29	0	100	38	9,966
Percent applications by a nurse - by NLC Status						
Within Home State (%)		39	0	100	43	9,966
Outside Home State - NLC to NLC (%)		32	0	100	42	9,966
Outside Home State - NLC to NonNLC (%)		24	0	100	37	9,966
Outside Home State - NonNLC to NLC (%)		7	0	100	24	9,966
Outside Home State - NonNLC to NonNLC (%)		9	0	100	26	9,966

Notes: Applications submitted through Wanderly between 2019H2 and 2023H2 (59,535 applications by 9,966 travel nurses). Source: Wanderly.

3.1 Descriptive Evidence

Appendix A3 presents figures illustrating how these patterns evolved over time in NLC and non-NLC states. Each figure plots a growth index normalized to 100 in 2019Q3. As Figure A6 shows, deaths per capita increased substantially more in non-NLC states than in NLC states during the first half of 2020. Thereafter, however, fluctuations in deaths per capita followed broadly similar patterns across the two groups of states. Although these raw differences are consistent with higher licensing barriers contributing to higher deaths per capita in non-NLC states at the onset of the pandemic, Figure A7 highlights an important confounding factor. Non-NLC states not only experienced a larger increase in deaths per capita but also faced a larger increase in COVID-19 cases per capita. Thus, a simple comparison of mortality between NLC and non-NLC states would conflate the effect of interstate licensing barriers on nurse supply with the fact that non-NLC states were also among the first and hardest hit by COVID-19.

Figures A8 and A9 show the trends in travel nurse applications and posted wages across NLC and non-NLC states, respectively. Applications to non-NLC states grew much more rapidly than applications to NLC states in 2020Q2. At first glance, this pattern may suggest that emergency licensing measures quickly relaxed the barriers facing nurses, allowing applications to respond without a substantial delay. However, as Figure A9 shows, posted wages also grew more rapidly in non-NLC states during 2020Q2. This pattern raises the question: if emergency measures fully eliminated the relevant licensing barriers, why did posted wages continue to exhibit differential growth across NLC and non-NLC states?

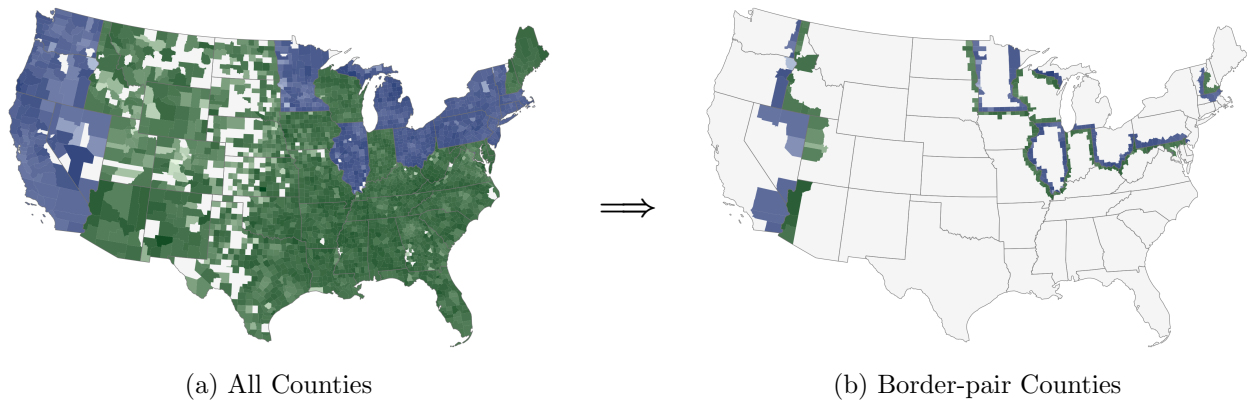
Figure A11 plots trends in the distribution of travel nurse applications over the full sample period. In 2019, there was a substantial gap between the share of applications to jobs where a nurse was licensed and the share to jobs where the nurse was not licensed. This gap narrowed after 2020 and remained small for the rest of the sample period. Because emergency measures temporarily relaxed interstate licensing requirements, this pattern suggests that nurses adjusted their application choices in response to these policy changes. However, the trends provide only suggestive evidence, since several other factors changed at the same time. The structural model in Section 5 separates these other channels from the effect of licensing barriers. Appendix A3 has the figures for all the graphs showing the different patterns in non-NLC versus NLC states over time.

4 Boundary Discontinuity Design (BDD)

To identify the effect of interstate licensing barriers on mortality, I exploit two sources of policy variation: cross-state differences in Nurse Licensure Compact (NLC) membership and the temporary suspension of licensing rules during the COVID-19 emergency. Specifically, I adapt a boundary discontinuity design (BDD) to isolate the effect of differences in barriers to interstate mobility between NLC and NonNLC states on all-cause mortality.

As shown in Table 1, NLC and NonNLC states differed along many dimensions beyond mortality and licensing barriers for RNs. Thus, a simple comparison of mortality between counties in NLC and non-NLC states would confound the effect of licensing barriers with these other differences. Using a BDD, I restrict the comparison to counties that share a state border, with one county located in an NLC state and the other in a non-NLC state (Figure 3). The balance checks below assess whether this approach reduces observable differences between NLC and non-NLC counties.

Figure 3: NLC and non-NLC Counties: All Counties and Border-pair Counties



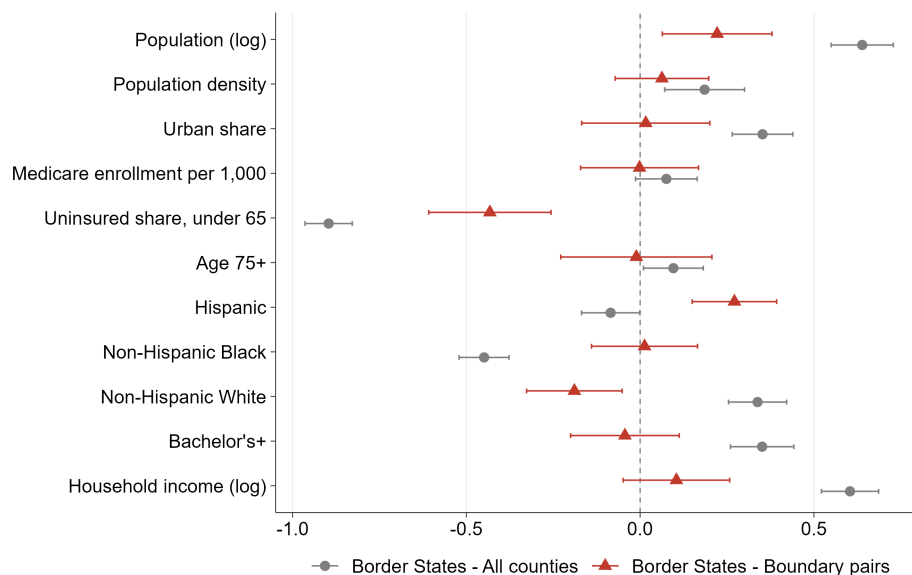
Notes: Counties are shaded by log all-cause deaths per capita in 2020, with non-NLC states in blue and NLC states in green. Panel (b) keeps only counties in cross-NLC border pairs. Source: CDC.

4.1 BDD: Balance Check

This section examines differences in the observable characteristics between NLC and non-NLC counties, comparing all counties in border states with counties that belong to a border pair. Figures 4 and 5 compare demographic characteristics and healthcare utilization during the baseline period, 2019H2. Each figure plots the standardized mean difference (non-NLC - NLC) for each baseline characteristic. Dots (gray) represent comparisons across all counties,

while triangles represent comparisons between counties on opposite sides of the same border pair.

Figure 4: Balance Check - Demographics Across NonNLC and NLC Counties



Notes: Each point is the standardized mean difference (non-NLC minus NLC, divided by the standard deviation) in a baseline characteristic in 2019H2, with 95% confidence intervals. Gray dots compare all counties in border states; red triangles compare the two sides of the same border pair.

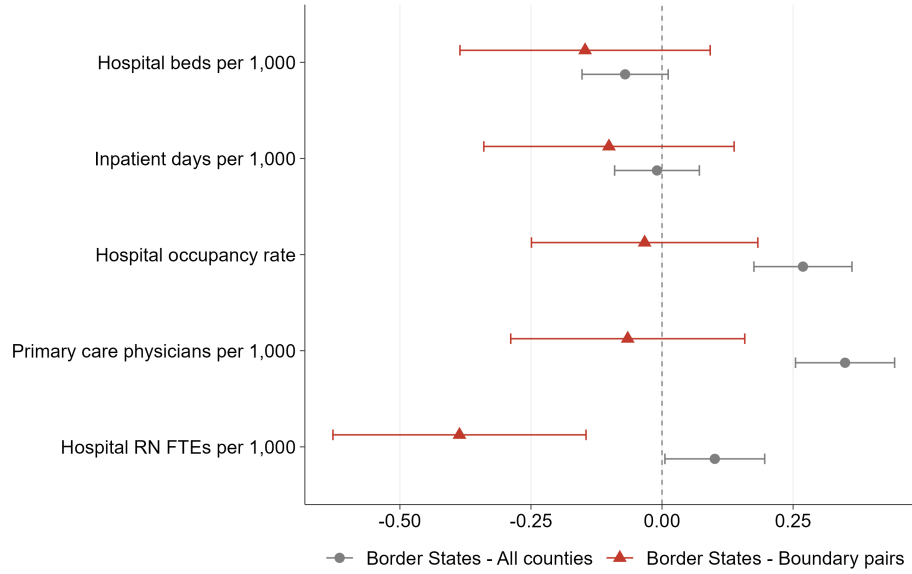
As Figure 4 shows, demographic differences in non-NLC versus NLC states narrow among border-pair counties, with most triangles closer to zero than the corresponding dots. A notable exception is the uninsured share of the population under age 65, which remains lower in non-NLC counties. This is a reflection of the broader differences in healthcare provision and insurance coverage across these states. This imbalance can confound the estimated relationship between licensing barriers and mortality, if greater insurance coverage increases healthcare utilization and consequently, mortality. However, Figure 5 shows no meaningful corresponding differences in healthcare utilization across border-pair counties.

As with demographics, differences in healthcare utilization between non-NLC and NLC counties narrow in the border-pair sample. A notable exception is the gap in RN FTEs per capita, which widens when comparison is restricted to border-pair counties. Non-NLC counties have fewer RNs per capita than NLC counties, consistent with differences in RN licensing barriers, even as other measures of healthcare utilization become more balanced.

Another potential confounder is the intensity of the COVID-19 outbreak. Although Figures 4 and 5 show that border-pair sample narrows difference in baseline demographics and healthcare utilization, the gray dots in Figure 6 indicate that non-NLC states initially

had substantially more COVID-19 cases per capita. Directly controlling for cases could introduce bad-control bias because caseloads may themselves be affected by licensing barriers. Moreover, no pre-emergency COVID-19 caseload exists for assessing baseline balance. The triangles in Figure 6, however, show that differences in cases per capita are close to zero among border-pair counties, including during the initial outbreak. By comparing neighboring counties across state borders, the BDD therefore balances their exposure to the COVID-19 shock and substantially reduces confounding from differences in outbreak intensity without having to directly control for caseloads in the regression.

Figure 5: Balance Check - Healthcare Utilization Across NonNLC and NLC Counties



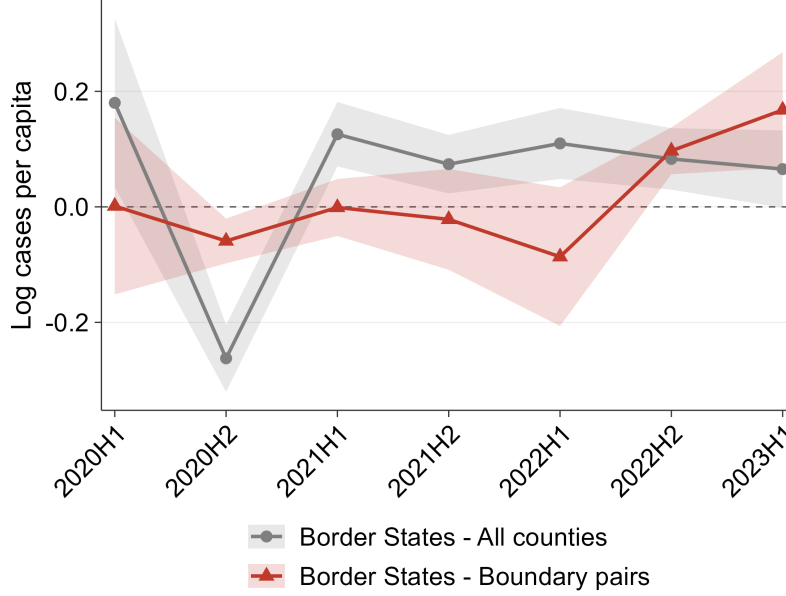
Notes: Each point is the standardized mean difference (non-NLC minus NLC, divided by the standard deviation) in a 2019 healthcare utilization measure, with 95% confidence intervals. Gray dots compare all counties in border states; red triangles compare the two sides of the same border pair. Source: AHRF.

4.2 BDD: Specification

To estimate the mortality effects of interstate licensing barriers for RNs, I estimate the following specification for county c in half-year t :

$$\log(y)_{ct} = \sum_{t \neq 2019H2} \beta_{1t} \{NonNLC\}_s \times \mathbb{1}\{Half-year\}_t + \beta_2 X_c + \beta_{(border-pair)t} + \epsilon_{ct} \quad (1)$$

Figure 6: Balance Check - COVID-19 Cases per Capita Across NonNLC and NLC Counties



Notes: Difference in log COVID-19 cases per capita between non-NLC and NLC counties by half-year, with confidence bands. Gray: all counties in border states; red: counties in border pairs.

where $y \in \{\text{deaths per capita}, \text{applications per capita}\}$.

The variable $\mathbb{1}\{NonNLC\}_s$ equals 1 if the state was not part of the NLC by January 2020, and 0 otherwise. The coefficient of interest is β_{1t} , which identifies how the effect of non-NLC status on y_{ct} changed over time, under the assumption of conditional exogeneity of $\mathbb{1}\{NonNLC\}_s$. To support this, the specification includes $\beta(border - pair)t$ and uses only neighboring counties that share a state border, with one county in an NLC state and the other in a NonNLC state. This reduces observable differences between the two groups, leaving the interstate licensing barrier for nurses as the main difference of interest. To account for any remaining differences, I add the baseline demographics control through X_c . Thus, the identifying assumption is that, after controlling for these factors, remaining differences between NLC and NonNLC border counties are due to differences in interstate licensing barriers for RNs.

First, I estimate Equation 1 for the *deaths per capita* with data from 2018-2024, and 2019H2 is the omitted period. The pre-2019H2 coefficients assess whether mortality in non-NLC and NLC counties followed similar trends before the pandemic. The post-period coefficients capture the differential effects of the COVID-19 shock (which is now equalized for the two groups, see Figure 6) on mortality across non-NLC and NLC counties due to dif-

ferences in interstate licensing barriers for RNs across these states. If licensing barriers were binding in non-NLC states, the sudden increase in demand for nurses should have increased mortality in non-NLC counties relative to NLC counties. However, if emergency measures effectively removed these barriers, this differential increase should disappear. Thus, a positive post-period estimate of β_{1t} would indicate that licensing barriers remained consequential despite the emergency measures, raising relative mortality in non-NLC counties.

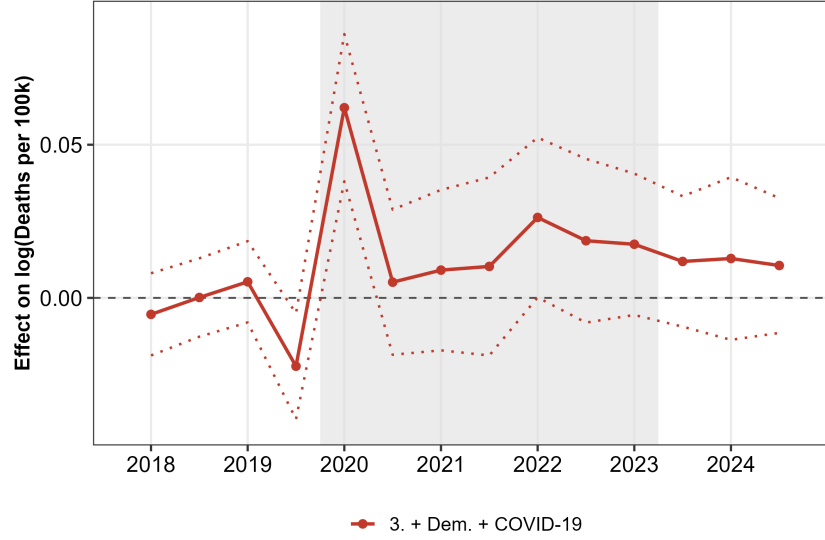
To provide suggestive evidence on the mechanism, I estimate Equation 1 for *applications per capita*, separately for the two professions: RNs and Allied. This comparison distinguishes an RN-specific response to licensing barriers from broader changes in healthcare labor supply. Wanderly data on travel nurse applications spans 2019H2-2023H2. Applications to boundary counties are sparse. Aggregating the data to the PUMA level provides some additional statistical power; however, the specification is still estimated on a thin sample and should be interpreted as suggestive evidence.

4.3 BDD: Results

The estimates from Equation 1 for *deaths per capita* are plotted in Figure 7. The coefficients for 2018 and 2019 are statistically insignificant, consistent with absence of differential pre-trends in mortality between non-NLC and NLC counties. The mortality gap increased sharply in 2020H1, when COVID-19 emerged and emergency licensing measures were introduced. Within the border-pair design, with counties similar on observables (including COVID-19 caseload), this spike indicates that higher licensing barriers initially constrained the RN supply response in non-NLC states, leading to higher mortality relative to NLC states. The subsequent closing of the gap suggests that the emergency measures eased these barriers, but with a lag. The 6% difference in $\log(\text{deaths per capita})$ in 2020H1 corresponds to 29 additional deaths per 100k during that time period. To examine the underlying mechanism and how the potential supply of RNs compared across non-NLC and NLC states, Figure 8 presents estimates for Equation 1 for RNs and Allied applications per capita.

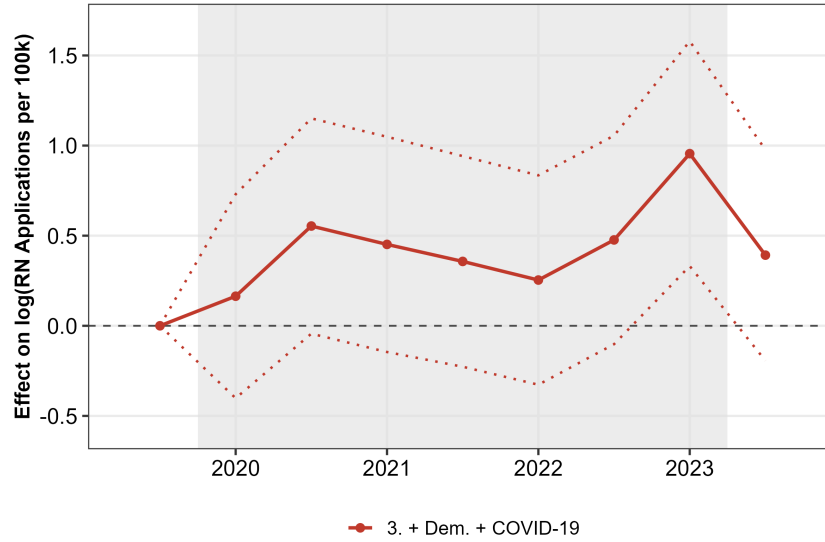
Because applications to border counties are sparse in 2019, these estimates have wider confidence intervals and limited statistical power. However, the point estimates reveal a pattern consistent with the mortality results. Non-NLC states received more RN applications per capita following the emergency measures, but this increase emerged only in 2020H2. This delayed increase coincides with the lagged closing of the mortality gap. The gap in applications per capita also shows a reversal once the emergency measures expired in 2023H1.

Figure 7: Coefficients β_{1t} for effect of $\mathbb{1}\{NonNLC\}_s$ on *deaths per capita*



Notes: Estimates of β_{1t} from Equation 1 for log deaths per 100,000, with demographic and COVID-19 controls and border-pair-by-half-year fixed effects. Coefficients are relative to the 2018H1–2019H1 average; dotted lines are 95% confidence intervals (standard errors clustered by border pair), and the shaded area marks the emergency period.

Figure 8: BDD Estimates: RN Applications per Capita (County-or-PUMA Units)



Notes: Estimates of β_{1t} from Equation 1 for log RN applications per 100,000, estimated on county-or-PUMA units with demographic and COVID-19 controls. Coefficients are relative to 2019H2; dotted lines are 95% confidence intervals (standard errors clustered by border pair).

5 Travel Nurse Applications: Model and Identification

In this section, I estimate a discrete choice model of travel nurse application decisions to quantify the effect of interstate licensing barriers on a nurse’s willingness to apply. A key feature of the Wanderly data is that they record every state license held by each applicant which allows to measure the licensing barrier at the nurse level. The licensing effect is identified from different application choices across nurses with different license holdings, for the same destination county in the same specialty. By allowing this effect to vary over time, the model traces how the licensing barrier changed as emergency measures temporarily suspended licensing requirements.

5.1 Supply: Travel Nurse Applications

Each application decision $i = 1, \dots, N$ in market $t = 1, \dots, T$ maximizes utility by choosing county c in the relevant specialty sp . I define the relevant choice set as $\mathcal{C}_{sp}$, the set of counties that receive at least one application in specialty sp during market t . The observed application is therefore modeled as a choice among counties active in the same specialty-market. The indirect utility of an application decision i from choosing a county c in market t is given by:

$$\begin{aligned}
 U_{ict} = & \sum_h \sum_k \beta_{1h}^k \mathbf{1}\{\text{NonLicensed}\}_{is(c)t}^k \times \mathbf{1}\{\text{Half-year}\}_{h(t)} \\
 & \underbrace{+ \beta_2 \log(\text{dist})_{ict} + \beta_3 \mathbf{1}\{\text{State-neighbor}\}_{is(c)t}}_{\mu_{ict}} \\
 & + \delta_{ct} + \epsilon_{ict}
 \end{aligned} \tag{2}$$

where the licensing category k of $\mathbf{1}\{\text{NonLicensed}\}_{is(c)t}^k$ is defined as

$$k = \begin{cases} 0 & \text{county } c \text{ is in nurse } i\text{'s home state,} \\ 1 & \text{outside home state; licensed through a multistate (NLC) license,} \\ 2 & \text{outside home state; licensed through a single-state license,} \\ 3 & \text{outside home state; not licensed in } s(c). \end{cases} \tag{3}$$

A simple comparison of licensed and unlicensed choices confounds the effect of licensing

barriers with nurses' destination preferences, as a nurse may hold a license in a particular state precisely because they prefer to work there. However, a multistate license grants nurses in NLC states access to all compact states, regardless of their preference for any particular state. I take that into account by assigning the counties in an application's choice set into one of the 4 categories as shown in Equation 3. With home state as the omitted baseline, $(\beta_{1h}^3 - \beta_{1h}^1)$ captures the effect of licensing barrier on application decisions and $(\beta_{1h}^2 - \beta_{1h}^1)$ captures the selection effect in every half-year.

Beyond licensing barriers, distance is an important factor affecting an application choice. The term $\log(distance)_{ict}$ controls for geographic distance between nurse i 's home state centroid and county c 's centroid. Together, these terms, denoted μ_{ict} , capture the application specific utility deviation from δ_{ct} , which represents the mean utility of county c in market t . The error term ϵ_{ict} is assumed to be Type I Extreme Value Distributed (EVD). The mean utility δ_{ct} can be further decomposed into observable and unobservable components:

$$\delta_{ct} = \delta_0 + \delta_1 \log(wage)_{ct} + \delta_2 \log(cases)_{ct} + \xi_{ct} \quad (4)$$

where $\log(wage)_{ct}$ is the log of RPP-adjusted hourly compensation posted by county c in week t , including the hourly equivalent of the housing stipend. To account for local COVID-19 severity in county mean utility, $\log(cases)_{ct}$ controls for COVID-19 cases per capita in county c .

A key concern in estimating the wage effect is that posted wages may be endogenous. Some counties may have lower wages because they provide better amenities. To address this concern, I instrument for Registered Nurse wages using wages posted for Allied positions in neighboring counties, with the following first-stage regression:

$$\log(wage)_{ct} = \alpha_c + \alpha_t + \alpha_1 \log(wage)_{nct}^{Allied} + \eta_{ct} \quad (5)$$

Taken together, Equations 2–5 form the supply side of the model, describing how nurses allocate applications across counties given licensing barriers, distance, and wages. In this framework, licensing barriers affect the application decision through two primary channels: the direct disutility of applying due to licensing barriers (β_1), and the endogenous wage premiums counties must offer to compensate for these barriers. The latter channel is captured by the wage coefficient δ_1 , which governs nurses' responsiveness to posted wages, together with the equilibrium wage adjustment implied by the county-level wage-posting model discussed next.

5.2 Demand: Wage Posting

Hospitals post vacancies and set wages to maximize their profits. Because the supply side is estimated at the county level, I model wage setting at the same level: all hospitals within a county are aggregated into a single representative employer. Following the approach of [Nevo \(2001\)](#), I use the county's wage-setting first order condition to back out the implied wage markdowns.

Let w_{ct} denote the wage posted by county c in market t and L_{ct} the number of travel nurses hired. Assuming decreasing returns to scale, county's profit function is given by:

$$\pi_{ct} = \underbrace{R(L_{ct}(w_{ct}))}_{\text{value-added}} - \underbrace{w_{ct} L_{ct}(w_{ct})}_{\text{wage bill}} \quad (6)$$

$$R(L_{ct}(w_{ct})) = \theta_{ct} (L_{ct}(w_{ct}))^{1-\alpha}, \quad \alpha \in [0, 1) \quad (7)$$

$$L_{ct}(w_{ct}) = \kappa A_{ct}(w_{ct}), \quad A_{ct}(w_{ct}) = N_t s_{ct}(w_{ct}) \quad (8)$$

Since I only observe applications and not the final hires in Wanderly data, I assume applications convert to actual hires L_{ct} at a proportional rate κ . Further, applications to a county can be written as the total applications in market t times the share of applications received by a county in market t - $s_{ct}(w_{ct})$. The wages w_{ct} must satisfy the first order condition:

$$\underbrace{R'(L_{ct})}_{\text{MRPL}} = w_{ct} \left(1 + \frac{1}{\epsilon_{ct}}\right) \Rightarrow \underbrace{w_{ct} = \text{MRPL}_{ct} \cdot \frac{\epsilon_{ct}}{1 + \epsilon_{ct}}}_{\text{Markdown Rule}} \quad (9)$$

where ϵ_{ct} is the elasticity of applications with respect to wages which can be expressed as:

$$\epsilon_{ct} = \frac{\partial \log s_{ct}}{\partial \log w_{ct}} = \delta_1 (1 - \bar{s}_{ct}), \quad \bar{s}_{ct} = \frac{\sum_i s_{ict}^2}{\sum_i s_{ict}} \quad (10)$$

Using estimates from the supply side, I can recover markdown estimates in the baseline without observing actual θ_{ct} . Assuming θ_{ct} does not get affected by licensing barrier, Equation 9 also allows to recover the wage effect of licensing barrier in the counterfactual with no barrier using baseline MRPL and ϵ_{ct} . Details are discussed in Section 6.

5.3 Identification

Licensing Effect: With home state as the omitted category, the model estimates 3 coefficients β_{1h}^k (for $k \in \{1,2,3\}$) for each half-year. For the same destination county during the same time period, nurses differ in whether they hold a license for that county's state. The coefficient is identified from whether nurses with and without a license choose a county at different rate, relative to other counties in their choice set. The model controls for other factors that can confound nurse's application choice: distance from the destination county and whether the destination county is in a neighboring state. Controlling for county-time fixed effect absorbs any other county specific factors in that time period that affected overall applications to that county. The difference $(\beta_{1h}^3 - \beta_{1h}^1)$ measures the licensing barrier in each half-year and $(\beta_{1h}^2 - \beta_{1h}^1)$ captures the selection effect of nurses into licenses.

Emergency measures adopted during COVID-19 temporarily removed the licensing barriers for interstate movement of healthcare workers in general, and RNs in particular. Since the model is estimated from 2019H2-2023H2, interacting the licensing barrier dummies allow to capture the effect of emergency waivers on the licensing effect on applications. This should affect only $(\beta_{1h}^3 - \beta_{1h}^1)$ and not $(\beta_{1h}^2 - \beta_{1h}^1)$.

Mean Utilities: Mean utility captures the factors that make a county attractive to all nurses, independent of nurse-specific characteristics. Wages and COVID-19 caseloads are the observed factors that may affect a county's mean utility in a given period. For a given market, the mean utilities are identified from differences in application shares across counties, after removing nurse-specific licensing and distance terms. The wage coefficient (δ_1) is identified from the exogenous change in RN wages over time within a county. Since wages are likely to be endogenous to the other unobserved county characteristics, the model uses allied wages as an instrument to identify (δ_1) .

5.4 Estimation

I estimate the supply-side model using the two-step approach of Berry, Levinsohn, and Pakes (1995). In the first step, I use maximum likelihood estimation (MLE) to estimate the parameters entering the individual-specific utility component, μ_{iht} . Given these estimates, I recover the mean utilities, δ_{ht} , using a contraction mapping algorithm that equates the model-predicted application shares to the observed shares for each ht pair.

The Type 1 extreme value distribution assumption on ϵ_{iht} implies that for a guess of β_1 , β_2 and δ_{ht} , the probability that application goes to hospital h in market t is given by:

$$P_{iht} = \frac{\exp(\mu_{iht} + \delta_{ht})}{\sum_{h' \in \mathcal{H}_{sp,t}} \exp(\mu_{ih't} + \delta_{h't})} \quad (11)$$

The log-likelihood function is given by:

$$\mathcal{L}(\beta, \delta) = \frac{1}{N \times T} \sum_{i=1}^N \sum_{t=1}^T \ln(P_{iht}(\beta, \delta)) \quad (12)$$

The first stage of estimation maximizes $\mathcal{L}(\beta, \delta)$, for a given starting value of β and δ . In the current version of the model, the choice set does not include an outside option of not applying in market t . As a result, the mean utilities δ_{ht} are identified only up to a market-specific normalization and are re-centered within each market during estimation.

In the second stage, I use the recovered mean utilities, δ_{ht} , from the first stage to estimate the parameters of the decomposition equation in 4. Because posted wages may be endogenous, Equation 5 provides an instrument for wages in 4. The second-stage parameters are therefore estimated using two-stage least squares (2SLS) method.

In the final step, I substitute the estimate of δ_1 from 9 to recover implied productivity, v_{ht} , using data on wages w_{ht} and application shares s_{ht} .

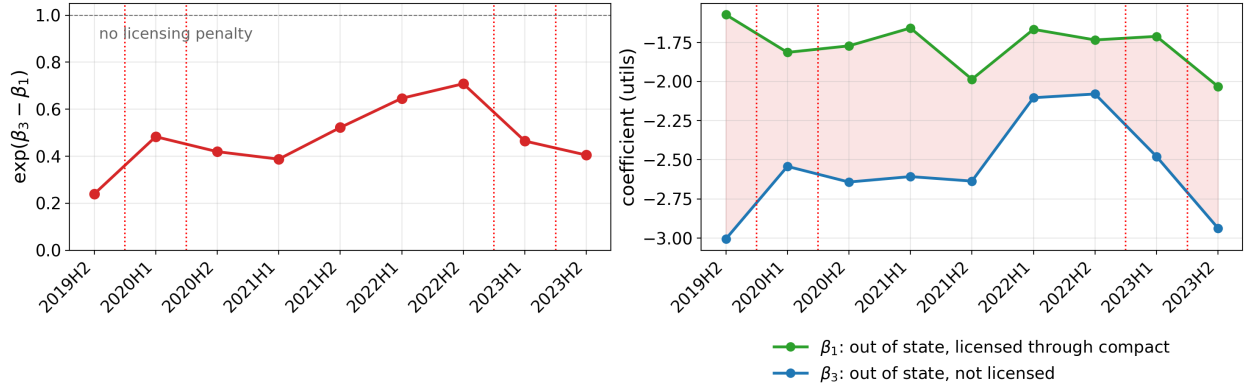
5.5 Estimation Results

5.5.1 Licensing Penalty over Time

The estimates of β_{1h}^3 and β_{1h}^1 are plotted in Figure 9. Their difference, $\beta_{1h}^3 - \beta_{1h}^1$, captures the licensing penalty in application decisions. The left panel plots the exponentiated difference, which should be below one if lacking a license deters applications. The estimates remain below one throughout the sample, indicating a persistent licensing penalty. However, the penalty declined after emergency measures were introduced in 2020H1 and began increasing again when they expired in 2023H1. The right panel shows that this pattern is driven primarily by changes in β_{1h}^3 , while β_{1h}^1 remains relatively stable. This supports interpreting the change as the effect of emergency licensing waivers, which directly altered the application choices captured by β_{1h}^3 . I next quantify the magnitude of this licensing penalty as nurses'

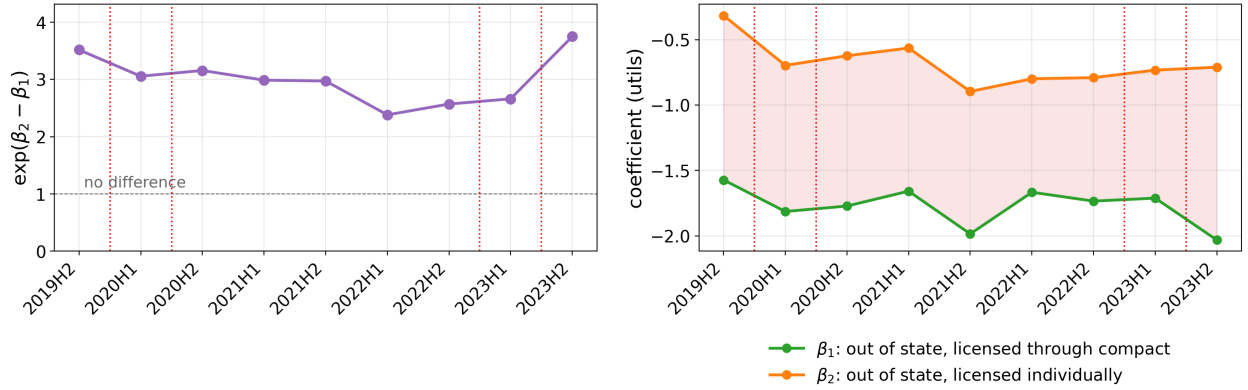
willingness to accept (WTA), using estimates of the wage elasticity of applications.

Figure 9: Licensing Penalty over Time: $\exp(\beta_{1h}^3 - \beta_{1h}^1)$ and Individual Coefficients



Notes: Left: $\exp(\beta_{1h}^3 - \beta_{1h}^1)$ by half-year, where a value of one means no licensing penalty. Right: β_{1h}^1 (licensed through the compact) and β_{1h}^3 (not licensed), relative to the home state; vertical dotted lines mark the start and end of the emergency period.

Figure 10: Selection over Time: $\exp(\beta_{1h}^2 - \beta_{1h}^1)$ and Individual Coefficients



Notes: Left: $\exp(\beta_{1h}^2 - \beta_{1h}^1)$ by half-year, where a value of one means no difference between individually held and compact licenses. Right: β_{1h}^1 (licensed through the compact) and β_{1h}^2 (licensed individually), relative to the home state; vertical dotted lines mark the start and end of the emergency period.

5.5.2 Wage Instruments and Elasticity of Applications

Table 3 presents the 2SLS estimates from Equations 4–5 using two alternative instruments for RN wages. With either instrument, the estimate of δ_1 is larger than its OLS counterpart. This is consistent with downward bias in OLS: counties with better amenities may attract applications while offering lower wages. The 2SLS estimates are similar across the two instruments, and the coefficient on log cases remains stable across specifications. Together,

these results support the robustness of the estimates. The wage coefficients imply that a 1% increase in wages raises applications by approximately 1.6%–2.1%.

Table 3: Mean utility on log wage and cases, equations (2)–(3)

Instrument	Allied IV	Agency IV
log(wage), OLS	1.1177	1.1071
log(wage), 2SLS (δ_1)	2.1146	1.5889
$f(\text{cases})$, OLS	0.0600	0.0588
$f(\text{cases})$, 2SLS	0.0347	0.0478
<i>First stage</i>		
Coefficient on instrument	0.1625	0.5955
Partial R^2	0.0296	0.0615
F-stat	132.61	81.22
Observations	9,429	10,091
County clusters	1,446	1,531

Notes: Estimates of Equation 4 with county and quarter fixed effects; the 2SLS rows instrument the log RN wage with the Allied IV or the Agency IV (Equation 5). The F-statistic is from the first stage, with standard errors clustered by county; observations are county-quarters.

The estimates for δ_1 and β_1 can be used to get the Willingness to Accept (WTA) for the licensing barrier for nurses using:

$$\text{WTA}_h = -\frac{\beta_{1h}^3 - \beta_{1h}^1}{\delta_1} \quad (13)$$

Table 4: Willingness to Accept for the Licensing Barrier (Allied IV)

Period	$\beta_{1h}^3 - \beta_{1h}^1$	WTA (log points)	WTA1 (%)
Pre-emergency (2019H2)	−1.43	0.68	97
During emergency (2020H1–2023H1)	−0.68	0.32	38
Post-emergency (2023H2)	−0.90	0.43	53

Notes: $\text{WTA}_h = -(\beta_{1h}^3 - \beta_{1h}^1)/\delta_1$ with $\delta_1 = 2.11$ (Allied IV). The emergency period reports the average over its half-years. $\text{WTA} (\%) = \exp(\text{WTA}) - 1$.

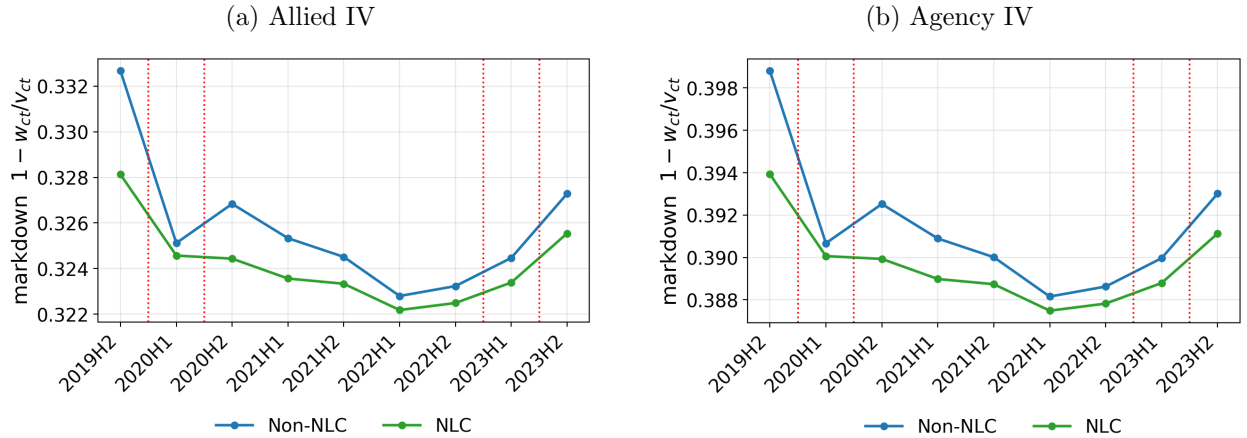
As Table 4 shows, pre-emergency, on average a nurse required 97% wage premium to be

willing to apply to a non-licensed position. This closed to 38% during emergency and started to go back up to 53% once emergency measures expired.

5.5.3 Markdowns

Using estimates from the supply side of travel nurse applications, I recover the marginal revenue product of labor (MRPL) from the markdown condition in Equation 9. Figure 11 plots the implied markdowns by the NLC status of the destination county. Higher licensing barriers make labor supply less elastic, allowing hospitals to impose larger markdowns. Thus, binding barriers should generate higher markdowns in non-NLC states, while emergency licensing measures should reduce them. Figure 11 shows both patterns. Markdown estimates are higher in non-NLC states throughout the sample, but decline in both groups after emergency measures are introduced in 2020H1 and begin rising again when the measures expire in 2023H1.

Figure 11: Wage Markdowns by NLC Status of the Destination State



Notes: Average markdown $1 - w_{ct}/v_{ct}$ by half-year for destination counties in non-NLC and NLC states, using δ_1 from the Allied IV in panel (a) and the Agency IV in panel (b). Vertical dotted lines mark the start and end of the emergency period.

6 Counterfactual

6.1 Counterfactual Estimation

The counterfactual of interest is the distribution of applications across locations in the absence of interstate licensing barriers. To construct this counterfactual, I eliminate the licensing disadvantage in utility by setting $\mathbb{1}\{NonLicensed\} = 1$ for all out-of-state hospitals in the choice set. Keeping everything else unchanged, this increases applications to hospitals in states with above-average exposure to licensing barriers and reduces applications to hospitals in states with below-average exposure. However, removing licensing barriers changes the supply of applicants available to hospitals and therefore their wage-setting incentives. The counterfactual must therefore allow wages to adjust endogenously rather than holding them fixed at their observed levels. I utilize Equation 9 to incorporate this. Equation 9 can be re-written as:

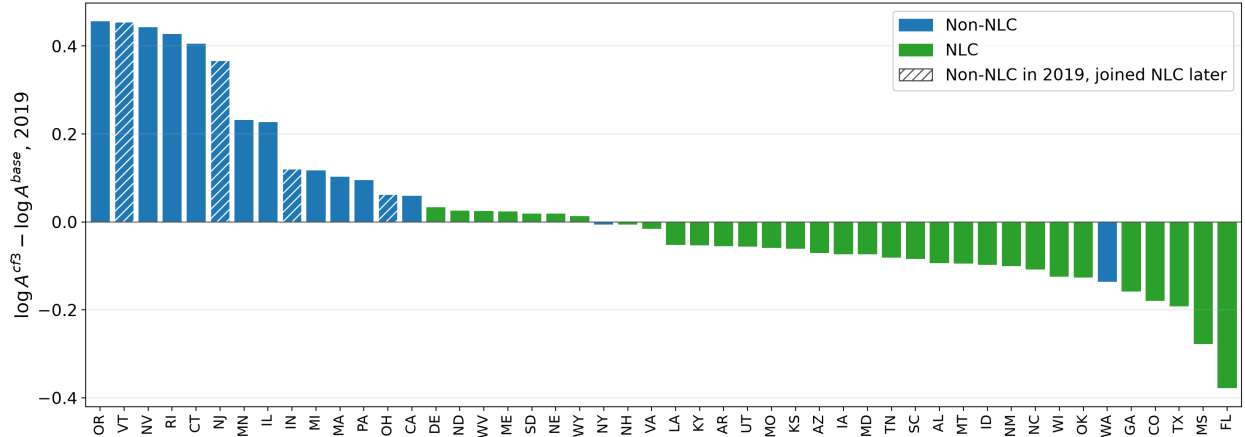
$$w_{ct} = \underbrace{v_{ct} \left(\frac{s_{ct}}{s_{ct}^0} \right)^{-\alpha}}_{MRPL_{ct}} \cdot \frac{\epsilon_{ct}}{1 + \epsilon_{ct}}, \quad \epsilon_{ct} = \delta_1 (1 - \bar{s}_{ct}) \quad (14)$$

Given estimates of v_{ct} and δ_1 , Equation 14 determines the equilibrium wages and application shares in the absence of licensing barriers. These barriers affect application supply through two channels: scarcity and elasticity. First, licensing barriers reduce the number of applications a county receives. Removing them therefore increases application supply and, through decreasing returns to scale, lowers wages - the scarcity channel. Second, licensing barriers restrict nurses' feasible choices, making application supply less elastic and increasing hospitals' wage-setting power. Removing them makes supply more elastic, reduces wage markdowns, and raises wages - the elasticity channel. I decompose the counterfactual wage change into these two opposing forces:

$$\underbrace{\Delta \log w_{ct}}_{\text{total}} = \underbrace{-\alpha \Delta \log s_{ct}}_{\text{(i) compensating differential}} + \underbrace{\Delta \log \frac{\epsilon_{ct}}{1 + \epsilon_{ct}}}_{\text{(ii) supply elasticity}}$$

After simulating the removal of the licensing barrier, I apply a dampened fixed-point iteration algorithm, iteratively updating nurse choice probabilities and optimal hospital wage responses until market shares and wages converge to a new steady state.

Figure 12: Change in applications across states in the absence of licensing barriers, 2019



Notes: The figure plots, for each state, the log difference between counterfactual applications with the licensing barrier removed and wages re-equilibrated, and baseline applications in 2019. Non-NLC states are in blue and NLC states in green; hatched bars are states that were not in the NLC in 2019 but joined later.

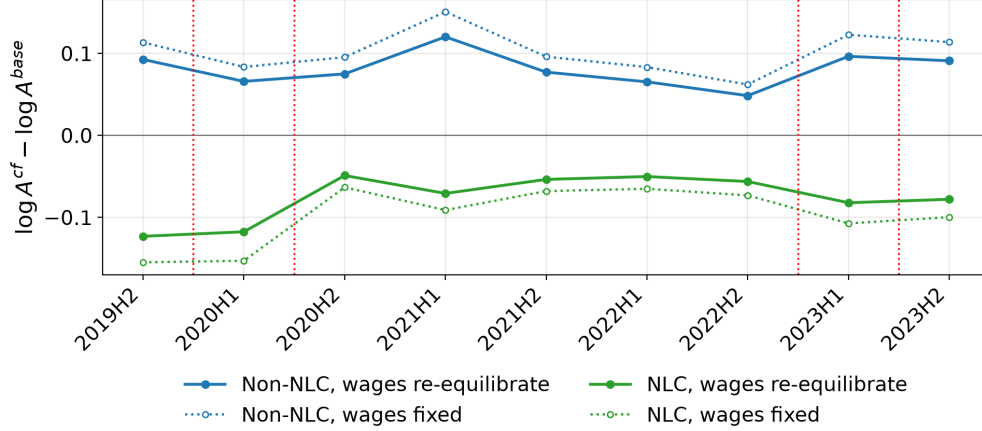
6.2 Counterfactual Results

In the counterfactual analysis, I shut down the licensing barrier for nurses by allowing all of them to hold the license for every state. The hospitals then re-equilibrate in response to supply changes. Figures 12, 13 and 14 show the results from counterfactual estimation.

Figure 12 ranks states by the average county-level percentage change in applications between the counterfactual and baseline for 2019. Although exposure to licensing barriers varies considerably across states, non-NLC states experience the largest application gains when the barriers are removed. Because licensing barriers in the model are defined using nurses' reported licenses rather than the NLC status of destination states, this pattern confirms that non-NLC states were, in practice, the most exposed to these barriers.

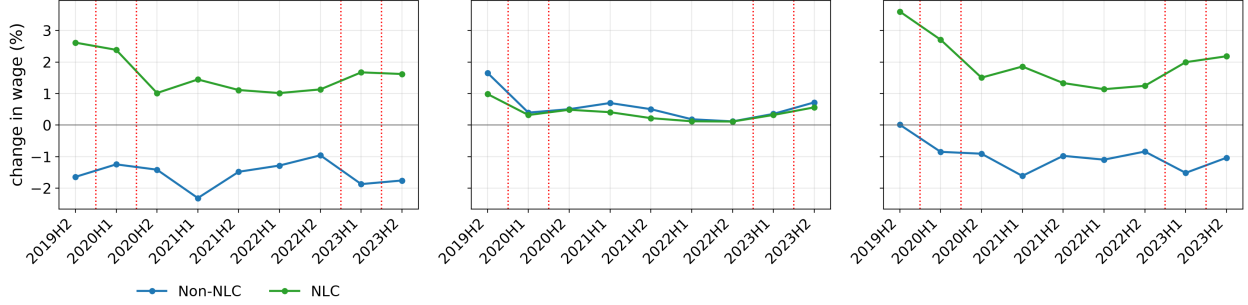
Figure 13 plots these counterfactual changes by half-year and NLC status. On average, removing licensing barriers would have increased applications to non-NLC states by 10% more than to NLC states. NLC states would also have gained applications through 2020H1, with their counterfactual gains disappearing only in 2020H2. This timing mirrors the lagged adjustment to the emergency measures observed in the reduced-form results.

Figure 13: Change in applications over time, Non-NLC vs NLC states



Notes: The figure plots the log difference between counterfactual and baseline applications by half-year, separately for Non-NLC and NLC states. Solid lines allow wages to re-equilibrate; dotted lines hold wages at their observed levels. Vertical lines mark the start and end of the emergency period.

Figure 14: Wage re-adjustment in the absence of licensing barriers, Non-NLC vs NLC states



Notes: The figure plots the percent change in wages by half-year, separately for Non-NLC and NLC states. Left panel: (i) compensating differential, $-\alpha \Delta \log s_{ct}$. Middle panel: (ii) supply elasticity, $\Delta \log[\epsilon_{ct}/(1 + \epsilon_{ct})]$. Right panel: total wage change, (i) + (ii).

6.3 Counterfactual Applications and Mortality

To link the counterfactual redistribution of applications to mortality, I construct for each county the change in its 2019 applications when the licensing barrier is removed and wages re-equilibrate,

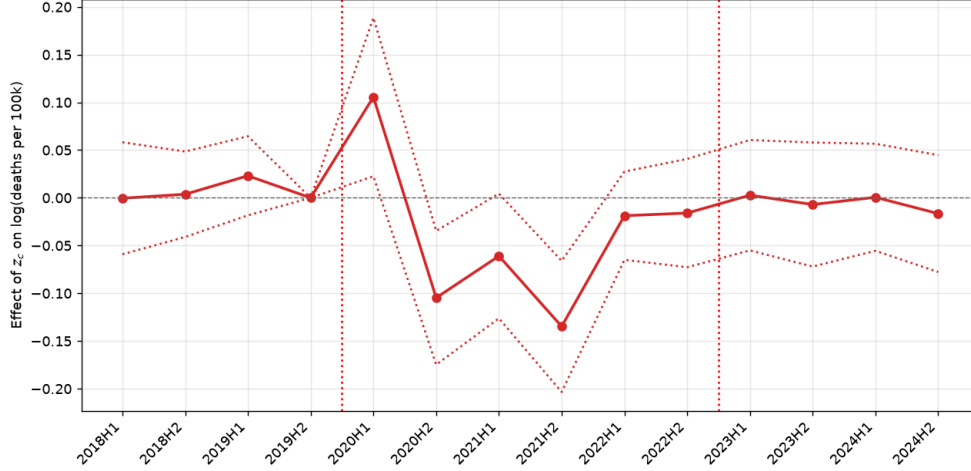
$$z_c = \log\left(\sum_{t \in 2019} A_{ct}^{cf}\right) - \log\left(\sum_{t \in 2019} A_{ct}^{base}\right).$$

A higher z_c indicates a county whose applications were held back more by the licensing barrier. I then estimate

$$\log(\text{deaths per capita})_{ct} = \sum_{t \neq 2019H2} \gamma_t z_c \times \mathbb{1}\{\text{Half-year}\}_t + \alpha_c + \tau_t + \varepsilon_{ct}, \quad (15)$$

where α_c and τ_t are county and half-year fixed effects. Figure 15 plots the estimates of γ_t . The coefficients are small and statistically insignificant before 2020. In 2020H1, counties more exposed to the licensing barrier experienced significantly higher mortality: a one standard deviation increase in z_c is associated with about 2% higher deaths per capita. The coefficients turn negative in 2020H2 and 2021H2 and return to zero from 2022 onward.

Figure 15: Counterfactual Change in Applications and Mortality



Notes: Estimates of γ_t from Equation 15 for 451 counties with 2019 applications and complete mortality data, 2018H1–2024H2, with 2019H2 as the omitted period. Dotted lines are 95% confidence intervals with standard errors clustered by state; vertical red lines mark the start and end of the emergency period.

7 Conclusion

This paper provides causal evidence that interstate variation in nurse licensing requirements constrained labor mobility and worsened early COVID-19 outcomes. Before the emergency waivers, a travel nurse required 96% wage premium to be willing to apply to a county she does not hold the license for. This gap decreased with emergency licensing waivers and went back up again once the waivers expired. The fact that the estimates closely track the timing of the emergency measures supports the interpretation that they capture the effects of licensing barriers.

The counterfactual estimates confirm that non-NLC states experienced the largest losses in applications because of their higher interstate licensing barriers. The timing of these effects shows that NLC states continued to receive more applications as a result of their lower licensing barriers through the first half of 2020. The difference in licensing-induced

application losses between non-NLC and NLC states closed only in the second half of 2020 and remained negligible thereafter, consistent with a delayed response to the emergency licensing measures. These, in turn, resulted in 6% higher deaths per capita in non-NLC states in 2020H1.

The results from this paper establish that interstate licensing barriers did affect the application decisions of travel nurses. While emergency waivers eased these barriers, it was with a lag and during a public health emergency such as COVID-19, this resulted in higher mortality.

The evidence also highlights the broader policy implications of uniform licensing frameworks like the Nurse Licensure Compact (NLC). States that had already adopted the NLC were better positioned to mobilize nurses during the initial surge of COVID-19, reflected in lower early mortality rates and faster labor market adjustment. Given the expanding scope of occupational licensing in the U.S., these results underscore that licensing mobility barriers can have serious consequences across professions, with particularly severe implications during emergencies when timely labor reallocation is essential.

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A1: Institutional Details

Figure A1: The Nurse Licensure Compact and Emergency Measures during COVID-19

NLC STORY

The Nurse Licensure Compact and COVID-19 – a Tale of Two States

POSTED 06/01/2020

Prior to COVID-19, New Jersey had enacted the NLC with plans for implementation later in 2020. Then on March 13th, as state governors were passing emergency declarations allowing nurses from outside their states to practice, New Jersey proceeded with an unprecedented partial implementation of the NLC, meaning nurses from out of state with the multistate license could immediately practice in New Jersey.

Because Missouri is part of the NLC, travel nurses from outside the state were able to remain there on staff during the pandemic.

"Having the extra staff on hand helped decrease nurse patient ratios during this time," says Kent.

The NLC also gave Missouri nurses the ability to leave the state, to help elsewhere. "If Missouri was not an NLC state, the nurses traveling

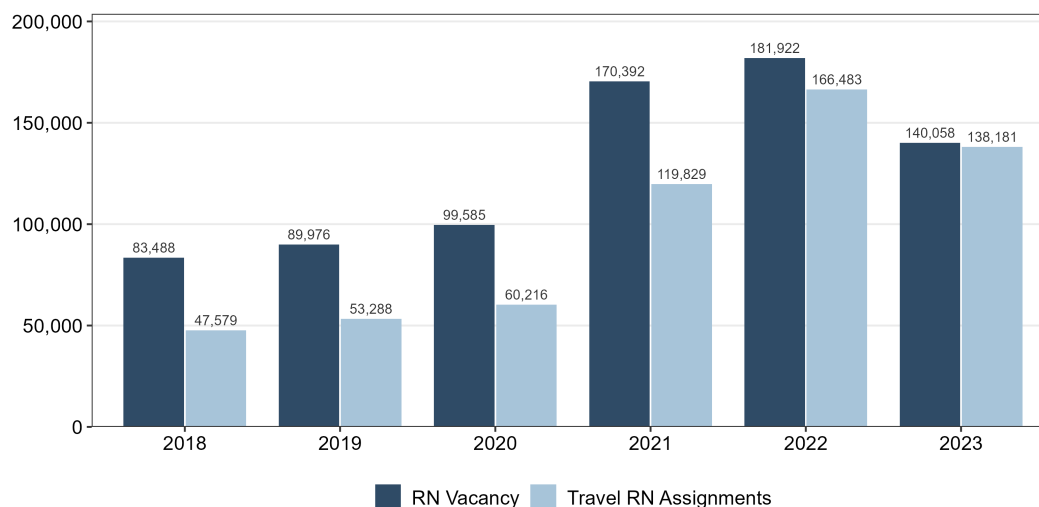
(a) New Jersey - Non-NLC

(b) Missouri - NLC

"I think the COVID-19 outbreak is going to cause the states that are not in the compact now to really take a second look at it," says NCSBN NLC Director Jim Puente, MS, MJ, CAE. "If the NLC was expanded to all 50 states, none of the guesswork with emergency orders would be necessary because nurses could travel to other states where they are needed. No applications, fees or background checks would be necessary. Disaster preparedness is a nonpartisan issue. The NLC is a solution to modernizing licensure and responding to disaster."

Notes: Source: Nurse Licensure Compact (NCSBN), "The Nurse Licensure Compact and COVID-19 – A Tale of Two States," posted June 1, 2020.

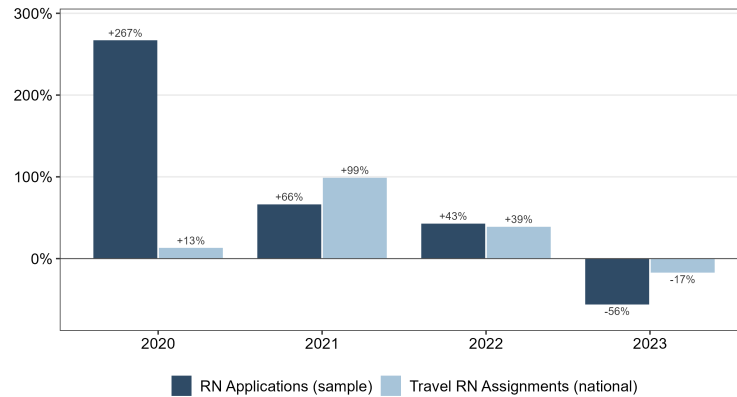
Figure A2: RN Vacancies and Travel RN Assignments, 2018–2023



Notes: RN vacancies are vacant hospital RN positions reported in the AHRF; travel RN assignments are the number of travel RNs on assignment nationally. Sources: AHRF; SIA/NATHO.

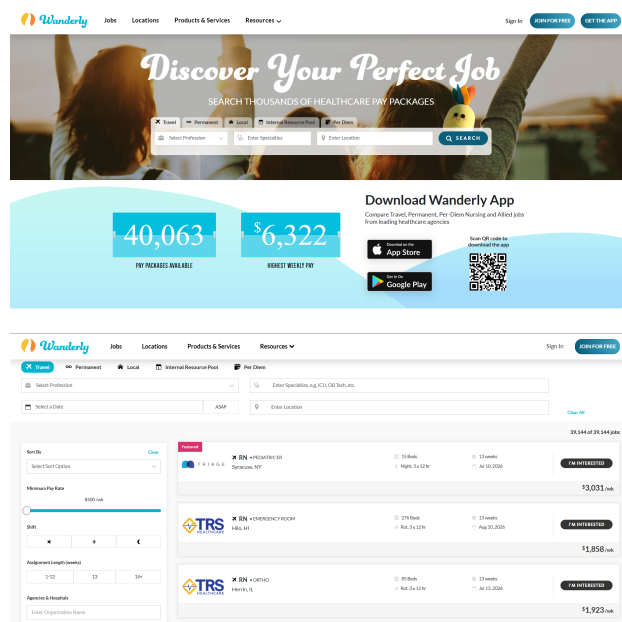
A2: Travel Nurse Data - Wanderly

Figure A3: Growth in Wanderly RN Applications vs National Travel RN Assignments



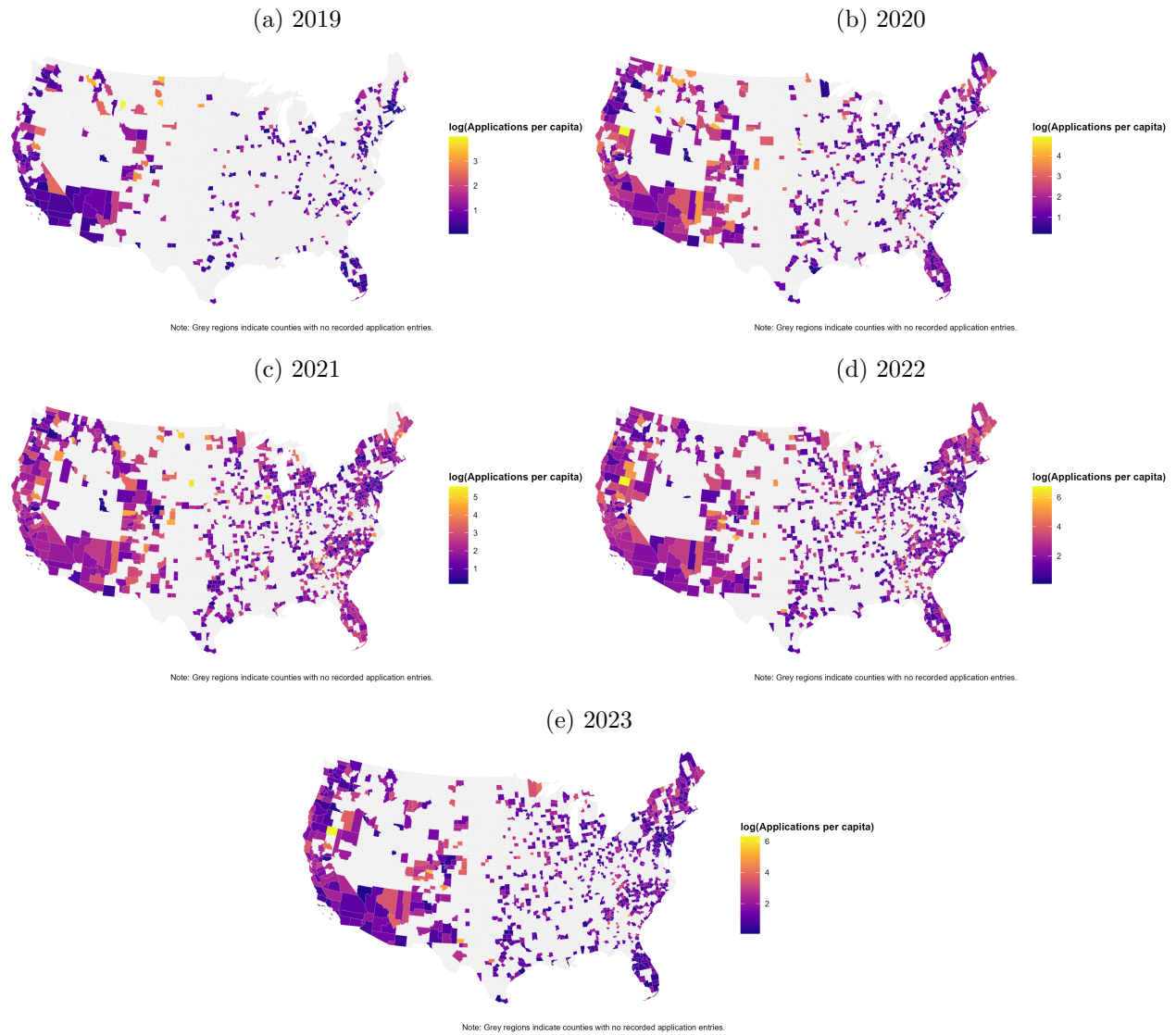
Notes: Year-over-year percent change in RN applications in the Wanderly sample and in national travel RN assignments. Sources: Wanderly; SIA/NATHO.

Figure A4: Snapshot of Wanderly Website



Notes: The figure above shows snapshot from the Wanderly website and illustrate what a travel nurse sees when choosing where to apply.

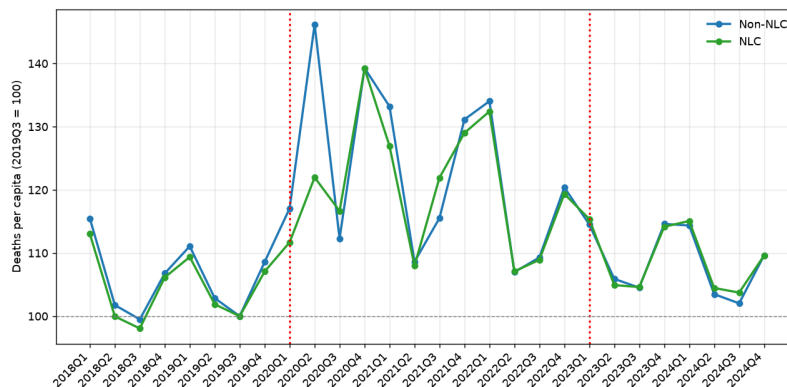
Figure A5: Distribution of Wanderly RN Applications per Capita by County, 2019–2023



Notes: Counties are shaded by log RN applications per capita received through Wanderly in each year; gray counties received no applications. Source: Wanderly.

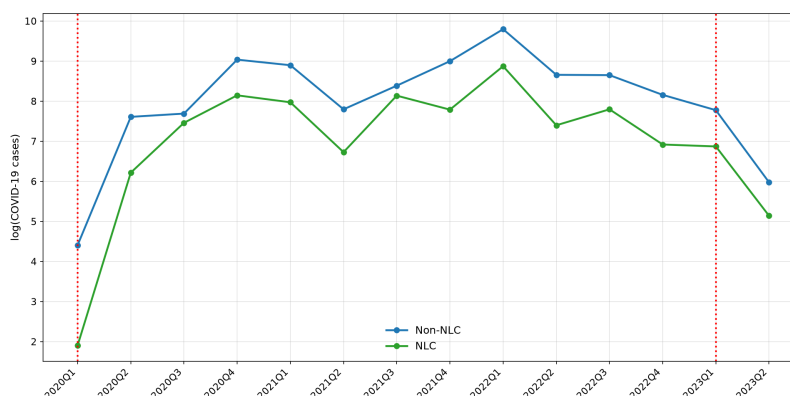
A3: Descriptive Evidence

Figure A6: Deaths per Capita in Non-NLC and NLC Counties, 2018Q1–2024Q4



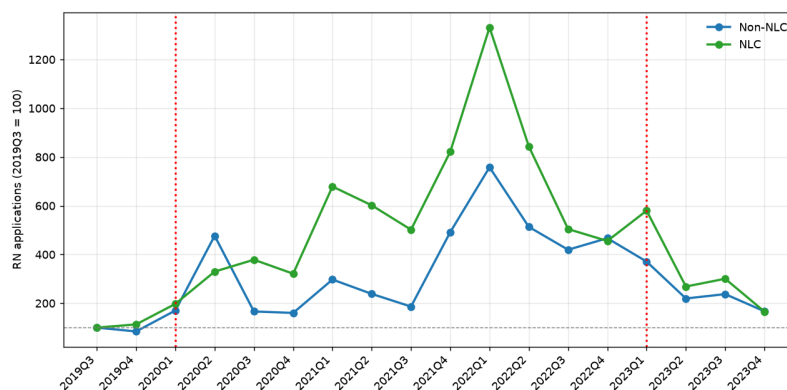
Notes: All-cause deaths per capita by quarter and NLC status as of January 2020, indexed to 2019Q3 = 100, for counties on either side of borders between NLC and non-NLC states. Red dotted lines mark 2020Q1 and 2023Q1. Source: CDC.

Figure A7: COVID-19 Cases in Non-NLC and NLC Counties, 2020Q1–2023Q2



Notes: Log of average COVID-19 cases per county by quarter and NLC status as of January 2020, for all counties. Red dotted lines mark 2020Q1 and 2023Q1. Source: CDC.

Figure A8: RN Applications to Non-NLC and NLC States, 2019Q3–2023Q4



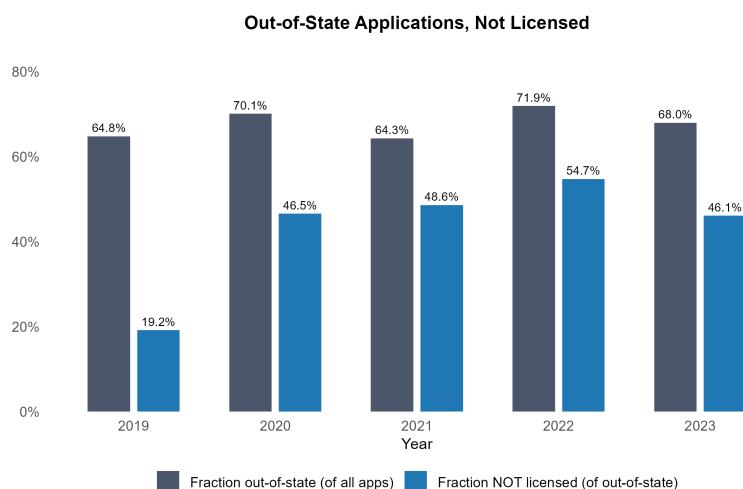
Notes: Number of RN applications to jobs in Non-NLC and NLC states by quarter, indexed to 2019Q3 = 100; states are classified by NLC status as of January 2020. Red dotted lines mark 2020Q1 and 2023Q1. Source: Wanderly.

Figure A9: Average Posted RN Wage in Non-NLC and NLC States, 2019Q3–2023Q4



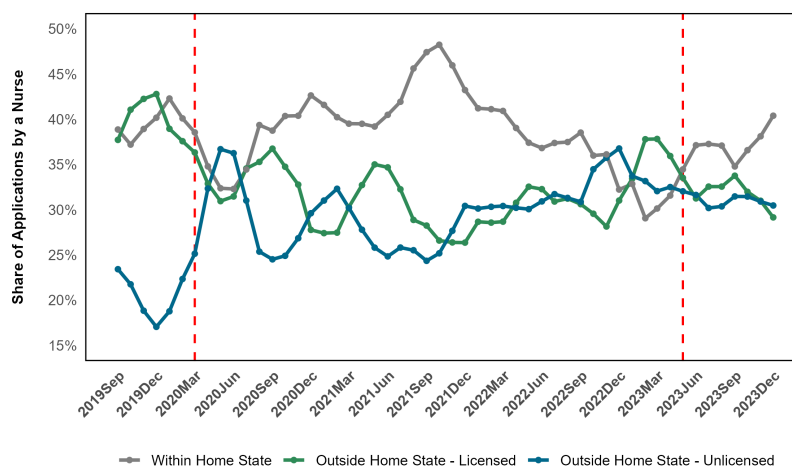
Notes: Average RPP-adjusted hourly loaded pay of RN job postings in Non-NLC and NLC states by quarter, indexed to 2019Q3 = 100; states are classified by NLC status as of January 2020. Red dotted lines mark 2020Q1 and 2023Q1. Source: Wanderly.

Figure A10: Out-of-State and Unlicensed Applications by Year



Notes: Dark bars show the share of all travel nurse applications made to jobs outside the nurse's home state. Blue bars show the share of these out-of-state applications made to jobs in states where the nurse does not hold a license. Source: Wanderly.

Figure A11: Share of licensed applications by a travel nurse



Notes: Monthly shares of travel nurse applications to jobs in the nurse's home state, to out-of-state jobs where the nurse holds a license, and to out-of-state jobs where the nurse does not. Vertical dashed lines mark the start and end of the emergency period. Source: Wanderly.

A4: Model Derivations

A4.1: Markdown First Order Condition

Each county c chooses its wage w_{ct} to maximize profits in Equation 6, taking the wages of all other counties as given:

$$\max_{w_{ct}} \pi_{ct} = R(L_{ct}(w_{ct})) - w_{ct} L_{ct}(w_{ct}).$$

Differentiating with respect to w_{ct} ,

$$\frac{\partial \pi_{ct}}{\partial w_{ct}} = R'(L_{ct}) \frac{\partial L_{ct}}{\partial w_{ct}} - L_{ct} - w_{ct} \frac{\partial L_{ct}}{\partial w_{ct}} = 0.$$

Rearranging,

$$R'(L_{ct}) = w_{ct} + L_{ct} \left(\frac{\partial L_{ct}}{\partial w_{ct}} \right)^{-1} = w_{ct} \left(1 + \frac{L_{ct}}{w_{ct}} \frac{\partial w_{ct}}{\partial L_{ct}} \right).$$

Since $L_{ct} = \kappa N_t s_{ct}(w_{ct})$, and neither the conversion rate κ nor the total number of applications N_t depends on w_{ct} (the choice set has no outside option),

$$\frac{w_{ct}}{L_{ct}} \frac{\partial L_{ct}}{\partial w_{ct}} = \frac{\partial \log L_{ct}}{\partial \log w_{ct}} = \frac{\partial \log s_{ct}}{\partial \log w_{ct}} \equiv \epsilon_{ct}.$$

Substituting into the rearranged condition above gives the first order condition in Equation 9:

$$\underbrace{R'(L_{ct})}_{\text{MRPL}_{ct}} = w_{ct} \left(1 + \frac{1}{\epsilon_{ct}} \right) \implies w_{ct} = \text{MRPL}_{ct} \cdot \frac{\epsilon_{ct}}{1 + \epsilon_{ct}}.$$

The county's share of applications in market t is the average of the individual choice probabilities,

$$s_{ct} = \frac{1}{N_t} \sum_i s_{ict}, \quad s_{ict} = \frac{\exp(\mu_{ict} + \delta_{ct})}{\sum_{c' \in \mathcal{C}_{sp}} \exp(\mu_{ic't} + \delta_{c't})}.$$

From Equation 4, $\partial \delta_{ct} / \partial \log w_{ct} = \delta_1$, and the logit form implies $\partial s_{ict} / \partial \delta_{ct} = s_{ict}(1 - s_{ict})$. Therefore,

$$\frac{\partial s_{ct}}{\partial \log w_{ct}} = \frac{1}{N_t} \sum_i \frac{\partial s_{ict}}{\partial \delta_{ct}} \frac{\partial \delta_{ct}}{\partial \log w_{ct}} = \frac{\delta_1}{N_t} \sum_i s_{ict}(1 - s_{ict}).$$

Dividing by $s_{ct} = \frac{1}{N_t} \sum_i s_{ict}$,

$$\epsilon_{ct} = \frac{\partial \log s_{ct}}{\partial \log w_{ct}} = \delta_1 \frac{\sum_i s_{ict}(1 - s_{ict})}{\sum_i s_{ict}} = \delta_1 \left(1 - \frac{\sum_i s_{ict}^2}{\sum_i s_{ict}} \right) = \delta_1 (1 - \bar{s}_{ct}).$$

Defining the markdown as the share of the marginal revenue product that is not paid to nurses,

$$1 - \frac{w_{ct}}{\text{MRPL}_{ct}} = 1 - \frac{\epsilon_{ct}}{1 + \epsilon_{ct}} = \frac{1}{1 + \epsilon_{ct}}.$$

Given the estimate of δ_1 and the observed wages and shares, the baseline MRPL is recovered as $v_{ct} = w_{ct}(1 + \epsilon_{ct})/\epsilon_{ct}$. The markdown falls as application supply becomes more elastic.

A4.2: Decomposition of the Counterfactual Change in Log Wages

With $R(L_{ct}) = \theta_{ct} L_{ct}^{1-\alpha}$,

$$R'(L_{ct}) = (1 - \alpha) \theta_{ct} L_{ct}^{-\alpha}.$$

Let superscript 0 denote the baseline equilibrium, so that $v_{ct} = R'(L_{ct}^0)$. Since $L_{ct} = \kappa N_t s_{ct}$, and θ_{ct} , κ and N_t are unchanged in the counterfactual,

$$\frac{R'(L_{ct})}{R'(L_{ct}^0)} = \left(\frac{L_{ct}}{L_{ct}^0} \right)^{-\alpha} = \left(\frac{\kappa N_t s_{ct}}{\kappa N_t s_{ct}^0} \right)^{-\alpha} = \left(\frac{s_{ct}}{s_{ct}^0} \right)^{-\alpha} \implies \text{MRPL}_{ct} = v_{ct} \left(\frac{s_{ct}}{s_{ct}^0} \right)^{-\alpha}.$$

Substituting into the markdown rule gives Equation 14:

$$w_{ct} = v_{ct} \left(\frac{s_{ct}}{s_{ct}^0} \right)^{-\alpha} \cdot \frac{\epsilon_{ct}}{1 + \epsilon_{ct}}.$$

Taking logs,

$$\log w_{ct} = \log v_{ct} - \alpha (\log s_{ct} - \log s_{ct}^0) + \log \frac{\epsilon_{ct}}{1 + \epsilon_{ct}}.$$

In the baseline, $s_{ct} = s_{ct}^0$, so

$$\log w_{ct}^0 = \log v_{ct} + \log \frac{\epsilon_{ct}^0}{1 + \epsilon_{ct}^0}.$$

Let superscript *cf* denote the counterfactual equilibrium without licensing barriers and

$\Delta x_{ct} = x_{ct}^{cf} - x_{ct}^0$. Subtracting the baseline from the counterfactual, $\log v_{ct}$ cancels:

$$\underbrace{\Delta \log w_{ct}}_{\text{total}} = \underbrace{-\alpha \Delta \log s_{ct}}_{\text{(i) compensating differential}} + \underbrace{\Delta \log \frac{\epsilon_{ct}}{1 + \epsilon_{ct}}}_{\text{(ii) supply elasticity}}, \quad \Delta \log s_{ct} = \log s_{ct}^{cf} - \log s_{ct}^0.$$

The decomposition is exact rather than a first-order approximation, because log wages are additively separable in the two terms. Both terms are evaluated at the counterfactual equilibrium, where s_{ct}^{cf} and $\epsilon_{ct}^{cf} = \delta_1(1 - \bar{s}_{ct}^{cf})$ solve the fixed point jointly with w_{ct}^{cf} .

Removing licensing barriers raises applications to the most exposed counties, $\Delta \log s_{ct} > 0$, so term (i) is negative: more applicants lower the MRPL under decreasing returns to scale, and wages fall. To the extent that removing barriers spreads a county's applications across more nurses, it lowers the share-weighted average $\bar{s}_{ct}$ and raises ϵ_{ct} . Since $\epsilon/(1 + \epsilon)$ is increasing in ϵ , term (ii) is positive: the markdown shrinks and wages rise. The sign of the total wage change depends on which channel dominates.

A5: Licensing Wage Premium: Reduced-Form Evidence

To examine whether jobs requiring a new license pay more, I compare the posted wages of out-of-state applications to jobs where the nurse is not licensed with those where the nurse is already licensed. For application i by nurse n to a job at hospital j in destination state s and quarter q of half-year h , I estimate

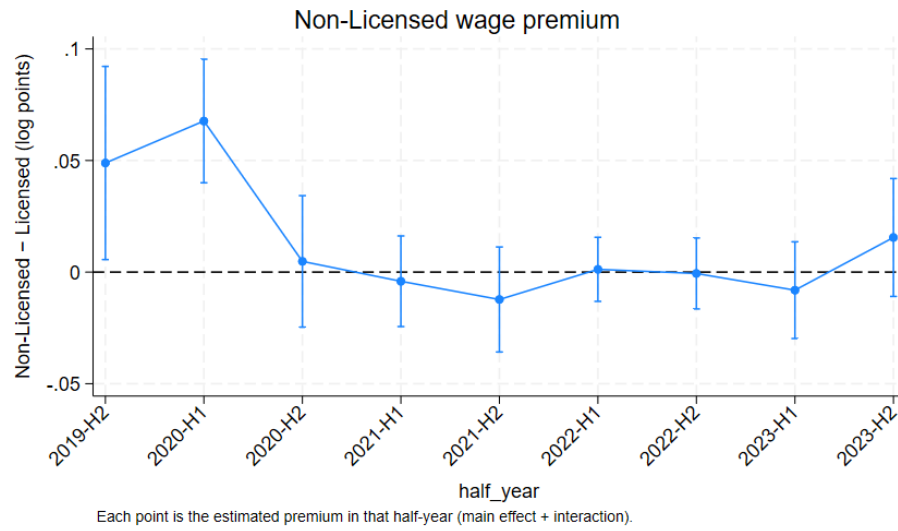
$$\begin{aligned} \log(wage)_{injsq} = & \sum_h \theta_h \mathbf{1}\{\text{NonLicensed}\}_{ins} \times \mathbf{1}\{\text{Half-year}\}_{h(q)} + \mathbf{X}'_{injq} \Gamma_{y(q)} \\ & + \alpha_q + \alpha_{z(j)} + \alpha_{u(i), y(q)} + \alpha_n + \lambda_s y(q) + \lambda'_s y(q)^2 + \varepsilon_{injsq}, \end{aligned} \quad (16)$$

where $\log(wage)_{injsq}$ is the log of the RPP-adjusted hourly compensation posted for the job, including the hourly equivalent of the housing stipend, and $\mathbf{1}\{\text{NonLicensed}\}_{ins}$ equals one if nurse n does not hold a license valid in state s . The vector $\mathbf{X}_{injq}$ contains a cubic in the standardized distance between the nurse's home state and the destination, a cubic in standardized county COVID-19 cases per capita, and the county shares of the population with at least one vaccine dose and with a completed primary series. Each control is interacted with year $y(q)$, so its effect can vary over time.

The specification includes quarter fixed effects α_q , hospital zip code fixed effects $\alpha_{z(j)}$, unit group-by-year fixed effects $\alpha_{u(i), y(q)}$, and nurse fixed effects α_n , as well as destination-state-specific linear and quadratic time trends $\lambda_s y(q) + \lambda'_s y(q)^2$. Nurse fixed effects mean that θ_h is identified by comparing the same nurse's applications to jobs with and without a valid license, and hospital fixed effects absorb persistent differences in pay across employers. The sample includes out-of-state RN applications from 2019Q3 onward. Standard errors are two-way clustered by destination state and nurse.

The coefficient θ_h measures the wage premium, in log points, for jobs in states where the nurse is not licensed relative to jobs in states where the nurse is licensed, in half-year h . Figure A12 plots these estimates.

Figure A12: Wage Premium for Applications to Jobs where the Nurse is Not Licensed



Notes: Estimates of θ_h from Equation 16: the difference in log posted wages between out-of-state applications to jobs where the nurse is not licensed and jobs where the nurse is licensed, by half-year. Bars are 95% confidence intervals, with standard errors two-way clustered by destination state and nurse. Source: Wanderly.